

Vedic Mathematics in Microcontrollers



Jai Sachith Paul worked at EFY till recently. The article is based on a talk given by P. Chow Reddy, managing director, Interleaved Technologies, at Electronics Rocks Conference 2014

When it comes to the embedded world, the microcontroller can perform a very large number of distinct functions. The performance of the controller for complex mathematical computations is limited by its speed. Digital signal processors (DSPs) are always an option for improved performance, but their use increases system complexity and cost of the project. Here comes the relevance of Vedic Mathematics. Logical calculations based on *sutras* in Vedic Mathematics can simplify computations, and significantly improve processing speed and performance of the controller.

Incorporating Vedic Mathematics into microcontrollers

In systems such as industrial process controllers, there are a good number of calculations to be done. Here, the process

should be controlled based on the output of these calculations. Therefore the time required by the microcontroller for these calculations is very critical.

We can consider a microcontroller as a clone of a human being in terms of calculations. Like a human being, a microcontroller can also perform calculations logically and in lesser time. Logics needed for these calculations are readily available from the *sutras*, which can be programmed into the application and made as a routine that can be called any time for calculating values. Based on logical decision, the controller selects which routine is to be called.

By using this technique, an increase in speed up to ten or even 15 times the original can be achieved. "A code written on 8051 series of microcontroller (using Vedic Mathematics) with 12MHz crystal to perform square of a two-digit number takes only 18 microseconds to generate an answer. Any other faster microcontroller, or even a PC, takes more than 200 microseconds in general mathematical methods. This almost eliminates the need of DSP microcontrollers," says P. Chow Reddy of Interleaved Technologies, Hyderabad.

Applying Vedic Mathematics

The idea can be adapted to various applications for making these faster and smoother. Let us take a look at some interesting applications.

What is Vedic Mathematics?

Vedic Mathematics refers to a set of 16 one-line mathematical formulae, or *sutras*, and their corollaries, rediscovered by a Hindu spiritual leader, Swami Bharati Krishna Tirtha from the *Vedas*—the ancient Indian scripture. It is a unique technique of logical calculations based on simple principles and rules with which any mathematical problem, arithmetic, algebra, geometry or trigonometry, can be solved mentally.

